

Interpretation of the 2025 Mathematical Model of the Contribution Protocol

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July 2026

1. Introduction

This paper does not revise or extend the mathematical model of the Contribution Protocol (CP).

Its purpose is to organize the 2025 mathematical model based on the designer's interpretation as of July 2026, without changing its original description, and to clarify the position of that interpretation.

2. Scope of This Paper

This paper concerns the mathematical model of CP published in 2025.

This paper organizes its contents from three perspectives: institutional design, the mathematical model, and interpretation.

In particular, the descriptions concerning π are treated in this paper not as mathematical theorems, but as the designer's interpretation.

This organization is also consistent with the appendix, "cp-limit-interpretation" (<https://github.com/contribution-protocol/contribution-protocol-project/blob/main/cp-limit-interpretation.md>), published in June 2026.

3. Institutional Design, Mathematical Model, and Interpretation

Mathematics is a language for describing an institution; it does not define the institution itself.

This paper organizes the 2025 mathematical model from the following three perspectives:

Institutional Design
Mathematical Model
Interpretation

Each of these has a distinct role and should be treated separately.

Institutional design is the highest-level concept that defines the purpose, principles, operational policies, and norms of the institution.

The mathematical model is a formal description for explaining the institution; it is not the institution itself. Therefore, the institution does not originate from mathematics. The institutional design existed first, and the mathematical model was constructed to describe its structure in a precise and formal manner.

Interpretation represents the designer's understanding of how the mathematical model should be understood and how its meaning and position should be regarded.

Interpretation may evolve through future research and practical implementation; however, this paper does not modify the mathematical model itself.

4. Current Interpretation of π

The mathematical model of CP was not intended to mathematically prove π itself.

The designer currently interprets π as the symbol that most closely represents, in numerical form, concepts such as love, prayer, and harmony, which cannot be fully expressed through mathematics alone.

Love is the starting point of the institutional design, and prayer arises from that love. Harmony is the direction toward which the institution aims over the long term. However, none of these can be rigorously defined within the mathematical model itself.

The mathematical model describes observable institutional behavior, including people's actions, records, institutional time, and institutional state.

5. The Institution's Journey Toward π

In CP, it is the institution itself that journeys toward π .

In this paper, the institution itself is understood as a sphere.

This sphere is not a static geometric figure; rather, it represents the institution as a dynamic entity that continually changes as institutional time progresses.

Institutional time advances through the accumulation of records within the institution. It is a concept distinct from chronological time.

Chronological time is physical time that passes at a constant rate, whereas institutional time advances only through recorded institutional activities. Therefore, even if chronological time passes, institutional time does not advance unless institutional activities are recorded. Conversely, if many institutional activities are recorded within a short period, institutional time also advances continuously.

In this sense, institutional time should be understood as a temporal axis based on institutional activities themselves.

The institutional state (z) is not the institution itself. Rather, it is an observable quantity derived from the accumulated records of the institution, and its state is understood by observing the institution over a certain period of time.

In CP, the history of the institution is preserved according to the append-only principle. Past records are neither deleted nor modified, and the institution continues while retaining its complete history.

Accordingly, the institution is an entity that continues to move through institutional time while encompassing its accumulated history from the past to the present.

The sphere continuously changes in response to institutional activities. Its size, shape, and internal state are not fixed, but continue to evolve through the operation of the institution and its ongoing practice.

Likewise, the six-directional vector is not the institution itself, but is understood as a cross-sectional observation of the sphere's state. Each component of the six-directional vector changes in response to institutional activities; however, each represents only one aspect of the sphere as a whole.

Therefore, in CP, it is neither the institutional state nor the six-directional vector that moves toward π , but the sphere as the institution itself.

The institutional state and the six-directional vector are regarded as observational indicators for observing that journey.

6. Conclusion

By distinguishing institutional design, the mathematical model, and interpretation, CP makes it possible to continue research while clearly defining the role of each.

This does not exclude the possibility that new mathematical models may be proposed in the future. However, such models will be positioned as independent research achievements separate from the 2025 mathematical model.

CP is not a system intended to reach conclusions within a short period of time.

It is expected that understanding of the protocol will deepen through long-term institutional operation and continuous observation.

DOI: (assigned upon publication)